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Model Analysis and Application of Quantum Computers in Credit Scoring Card Combination Optimization

Xiaohan Zhou [1], Ruikun Zhang[1*]

1. School of Mathematics and Physics, Qingdao University of Science and Technology, Qingdao 266061, China

*Corresponding author: rkzhang@qust.edu.cn

Abstract: In the context of business such as bank credit cards or related loans, banks score customers through credit scoring cards. Loan interest income and bad debt losses determined by the combination of thresholds of different credit scoring cards are key factors that determine the bank's final income. This paper mainly studies the combination optimization of credit scoring cards by establishing an optimization model and transforming it into a Quadratic Unconstrained Binary Optimization (QUBO) problem. First, the optimization problem of a single credit scoring card is studied, the data is preprocessed and zero-one integer programming model is established, then convert the objective function into a Hamiltonian to obtain an optimization model in the form of QUBO, and obtain the maximum bank income and the threshold of a single credit scoring card through the Simulated Annealing. Then, the credit scoring card combination optimization problem is studied. On the basis of the above research, variable substitution is used to reduce function degree, and a penalty function is introduced to obtain an unconstrained QUBO model for the combination problem. The D-Wave quantum annealing algorithm is used to obtain the optimal combination solution.

Keywords: QUBO, Zero-one Programming, Hamiltonian, Penalty Function, Quantum Annealing Algorithm, Credit Scoring Card

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1 Introduction

With the continuous development of the financial sector since the 20th century, the credit scoring card has gradually evolved into a pivotal tool in banking. This tool evaluates a customer's creditworthiness in an objective and systematic way by considering multiple factors related to their creditworthiness [1]. The design and application of the credit scoring card play a vital role in the risk management and revenue optimization of banks in credit operations.

The design of the credit scoring card is related to the number of customers who pass the loan qualification review and the interest income earned by the bank. For banks, finding a balance point by reasonably setting the threshold of the credit scoring card can reduce the risk of non-performing loans and ensure that the risk is controllable, and achieve maximum profitability to better suit market needs and the bank's strategic goals.

In order to explore the credit scoring card optimization and portfolio problem, many scholar teams have conducted research in this field. In [2], the impact of loan interest rates and banking market structure based on venture capital requirements is explained, reflecting the importance of reducing default. In [3], a modeling study is conducted on the evolution of bank typified loan portfolios to investigate bank credit losses, and the impact of expected loss models on the periodicity and capital ratio of bank credit losses is evaluated using practical cases. In [4], with the aim of maximizing the net profit of a credit card company, the article identifies customers with the same credit limit and genetic algorithms are used to find the optimal credit scoring card adjusted credit limit for customers. In [5], the semi Markov process method is used to optimize the investment portfolio composed of credit risk bonds, achieving dynamic credit rating and finding the optimal combination to minimize overall credit risk.

However, the above studies mainly use methods such as machine learning for optimization evaluation, whereas the QUBO model has the advantages of quantum computing, which can solve optimization problems more efficiently and comprehensively. It can run on quantum computer hardware and perform millisecond-level acceleration through quantum computers. It has been successfully researched and applied in the fields of computer [6] and physics [7]. This model and acceleration method will be widely used in various industries in the future. With the emergence of various types of quantum processors, the transformation and derivation of quantum heuristic algorithms are becoming more and more important. In [8], the application of quantum models and quantum algorithms in optimization problems are introduced in detail, achieving the alternation of multiple types of algorithms between a common series of operators. In [9], by introducing an Ising model equivalent to the QUBO model, various NP problems have been solved, and ideas have been provided for finding optimization problems that satisfy certain objectives. Therefore, how to establish a suitable model, convert the bank credit scoring card combination optimization problem into QUBO form, and determine the best combination method based on the model is an issue of important research value, which is also the main research motivation of this article.

This paper focuses on the application of using Quadratic Unconstrained Binary Optimization model in credit scoring card combination optimization. According to the characteristics and functions of the QUBO model, a quadratic unconstrained binary optimization model is established and solved by the quantum annealing algorithm to find one credit scoring card and its corresponding threshold and three credit scoring cards and their threshold combinations among 100 credit scoring cards respectively, so that the bank will have the most revenue in the end.

The contributions can be summarized as follows:

(1) A single credit scoring card threshold optimization model in the form of QUBO is developed, which is solved by a simulated annealing algorithm to make the bank end up with the most revenue.

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(2) A threshold combination optimization model for credit scoring card in the form of QUBO is developed and solved using the D-Wave quantum annealing algorithm.

The article is organized as follows: Section 2 presents the single credit scoring card optimization problem. Section 3 solves the credit scoring card portfolio optimization problem by establishing an optimization model in the form of QUBO. Section 4 is the conclusion of the article.

2 Single Credit Scoring Card Optimization

Now there are 100 credit scoring cards and their corresponding pass rate and bad debt rate data, each card can be set one of the 10 thresholds, to find out 1 out of 100 credit scoring cards and their corresponding thresholds to maximize the bank's final income. First of all, the bank scenario process is simplified [2], assuming that under the premise of loan funds of 1 million yuan, the bank loan interest income rate of 8%, the bank's final income is determined by two parts of the loan interest income and bad debt loss, the loan interest income is: loan funds × interest income rate× total pass rate ×(1 - total bad debt rate), bad debt loss is: loan funds× total pass rate× total bad debt rate.

Analyze the pass rate and bad debt rate corresponding to the credit scoring card thresholds, study the correlation between the two, and draw a line chart of the pass rate and bad debt rate corresponding to each threshold of 100 credit scoring cards.

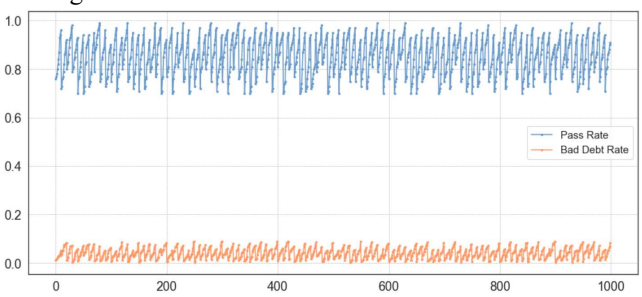

Fig. 1:Line chart of pass rate and bad debt rate

As shown in Fig. 1, we can find that the data changing trends of the bad debt rate and the pass rate are basically the same. Generally, the higher the pass rate, the higher the bad debt rate will be. There is a positive correlation between the above two. And the fluctuation range of its pass rate is roughly between 0.7-0.9, while the fluctuation range of bad debt rate is roughly between 0-0.15.

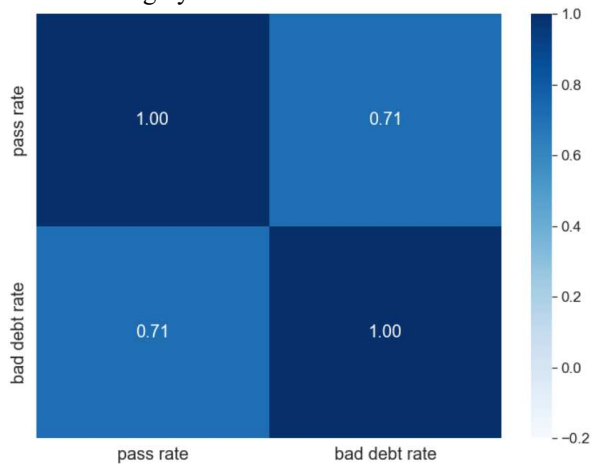

Fig. 2:Correlation heat map of pass rate and bad debt rate

By calculating the Pearson correlation coefficient as shown in Fig. 2, it can be found that the pass rate and the bad debt rate have a strong positive correlation, and the correlation coefficient reaches 0.71. Therefore, banks need to balance the pass rate and the bad debt rate when formulating the credit scoring card threshold. Improving pass rates may help business growth, but may also lead to higher bad debt rates.

First, the data is preprocessed and coded, introduce the binary variable x_i ,and define the coding method as follows.

If :

$$\begin{cases} m=[i/10], m=1,2,...,100 \\ n=[i\%10], n=0,1,...,9 \end{cases} \tag{1}$$

where $[x]$ is a ceiling function, we specify that when $n=0$, the threshold is set to 10, x_i indicates whether to select the n-th threshold of the m-th credit scoring card, and $x_i \in \{0,1\}, i=1,2,...,1000$.

Hence, interest income on loans:

$$W_1 = 100\times 0.08\times a_i \times (1-b_i) \tag{2}$$

Bad debt losses:

$$W_2 = 100\times a_i \times b_i \tag{3}$$

Final bank income:

$$W = W_1 - W_2 = 8a_i - 108a_i b_i \tag{4}$$

where a_i, b_i are the pass rate and bad debt rate for the credit scoring card for which the threshold is determined.

The objective is determined to be the maximization of the bank's ultimate income, which consists of two components, interest income on loans and losses on bad debts, so the objective function can be expressed as:

$$\begin{aligned} &maxW = \sum_{i=1}^{1000}(8a_i - 108a_i b_i)x_i \\ &s.t.\begin{cases} \sum_{i=1}^{1000} x_i = 1 \\ x_i \in \{0,1\}, i=1,2,...,1000 \end{cases} \end{aligned} \tag{5}$$

Due to the wide variety of combination problems, although brute force enumeration can even obtain relatively accurate results when the amount of data is small, it is computationally complex and time-consuming. Moreover, when the amount of data is huge, the computing power, time and space complexity occupied by violent enumeration are far higher than the existing normal computer computing power level. Therefore, for combination problems, the optimization model is converted into a QUBO model. Solving will greatly shorten the calculation time, and we can obtain the results we need efficiently and more accurately.

Next the model is transformed into the QUBO form [10], which is a mathematical model for solving combination optimization problems. In the QUBO model, the problem needs to be transformed into an optimization model in which the decision variable is a binary variable and the objective function is a quadratic function.

The general form of QUBO is:

$$H(x) = \sum_i q_{ii}x_i + \sum_{i<j} q_{ij}x_i x_j \tag{6}$$

where $H(x)$ is the objective function of QUBO, it repre-

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sents the Hamiltonian quantity or energy function of the problem; x_i is a binary variable that can take the value of 0 or 1; q_{ii} is a linear coefficient associated with a single variable x_i ; and q_{ij} is a quadratic coefficient associated with two different variables, x_i and x_j , $i \neq j$; that is, the first term of the QUBO model is the sum of the linear terms, and the second term is the sum of the quadratic terms, and the goal is to find the binary variable x , that minimizes the objective function $H(x)$.

At the same time, the QUBO model can be represented as a matrix form of quadratic terms. For a N dimensional binary variable $X = (x_1\ x_2 \cdots x_N)$ corresponding to a decision variable N , the objective function of the QUBO model can be represented by a symmetric matrix of $N \times N$, where the elements Q_{ij} of Q is a coefficient associated with the variables x_i and x_j .

The matrix Q is a constant square matrix, usually assumed to be symmetric or upper triangular. If Q is converted into a symmetric matrix, all elements Q_{ij} except $i = j$ need to be converted to $\frac{Q_{ij} + Q_{ji}}{2}$; if Q is converted into an upper triangular matrix, all elements Q_{ij} with $j > i$ need to be converted to $Q_{ij} + Q_{ji}$, and then the elements $j < i$ is 0, if the matrix is symmetrical, you need to double the element values above the main diagonal, and then set all values below the main diagonal to 0.

Therefore, the objective function $H(x)$ can be written in matrix form: $H(x) = X^T QX$ .The diagonal elements of the matrix Q , Q_{ii} correspond to linear terms and the non-diagonal elements Q_{ij} correspond to quadratic terms.

In this paper, the decision variable "credit scoring card and its corresponding threshold" is transformed into a binary variable and the objective function is transformed into a quadratic function. Since QUBO usually seeks to minimize the objective function to solve the problem, a new coefficient $q_i = -(8a_i - 108a_i b_i)$ is defined for the problem, and since x_i takes the value of 0 or 1, the coefficient $x_i = x_i^2$ is defined for the problem.

Thus, the objective function that maximizes the bank's income is obtained:

$$maxW = min(-W) = \sum_{i=1}^{1000} q_i x_i^2 \tag{7}$$

The solution is further solved using the matrix form, which transforms the objective function into the form of $H(x) = X^T QX$.

Define the X-vector as $X = (x_1, x_2, ..., x_{1000})^T$, we get the Hamiltonian operator $H(x) = \sum_{i=1}^{1000}\sum_{j=1}^{1000} q_{ij}(1-x_i)(1-x_j)$, the coefficients q_{ij} are related to credit scoring card pass rates and bad debt rates, and

$$q_{ij} = -2q_i\delta_{ij} + 2q_i q_j,\ \delta_{ij} = \begin{cases} 1, i = j \\ 0, i \neq j \end{cases} \tag{8}$$

The form of matrix Q is:

$$Q = \begin{pmatrix} q_{11} & 0 & \dots & 0 \\ 0 & q_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & q_{1000,1000} \end{pmatrix}$$

$$= \begin{pmatrix} \begin{matrix} -8a_1 + \\ 108a_1b_1 \end{matrix} & 0 & \dots & 0 \\ 0 & \begin{matrix} -8a_2 + \\ 108a_2b_2 \end{matrix} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \begin{matrix} -8a_{1000} + \\ 108a_{1000}b_{1000} \end{matrix} \end{pmatrix} \tag{9}$$

Thus the rectangular form of the objective function is obtained as:

$$X^T QX = \begin{pmatrix} x_1 & x_2 & x_3 & \cdots & x_{1000} \end{pmatrix}$$

$$\begin{pmatrix} \begin{matrix} -8a_1 + \\ 108a_1b_1 \end{matrix} & 0 & \dots & 0 \\ 0 & \begin{matrix} -8a_2 + \\ 108a_2b_2 \end{matrix} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \begin{matrix} -8a_{1000} + \\ 108a_{1000}b_{1000} \end{matrix} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{1000} \end{pmatrix} \tag{10}$$

Since it is a small scale optimization solution, typical optimization algorithms can be used to solve the QUBO problem. Simulated Annealing Algorithm [11] is a probabilistic based global optimization algorithm for the problem of finding a globally optimal solution in a complex space.

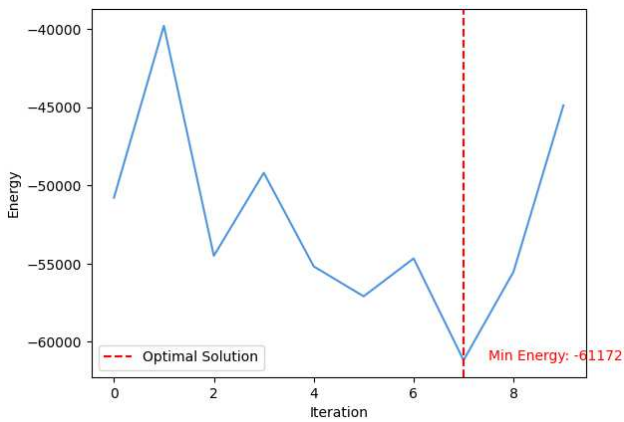

Fig. 3:Simulated annealing process

Firstly, initialization is performed to randomly generate the initial solution, that is, the selection of 100 credit scoring cards and the corresponding thresholds, then the solution x_i is mapped as a variable in the QUBO model, defining the number of energies as $H(x)$, searching for a new solution within the neighborhood of the current solution, generating the solution by adjusting the thresholds for the

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credit scoring cards, deciding whether or not to accept the new solution based on the Metropolis criterion, and implementing the temperature dispatch strategy, which gradually reduces the temperature. Multiple iterations of the above process gradually approach the globally optimal solution.

It can be seen from the energy change diagram Fig. 3 during the simulated annealing process that as the iteration proceeds, the energy value generally shows a downward trend, and the point where the energy reaches the minimum value corresponds to the optimal solution of 61,172 CNY found by the simulated annealing algorithm.

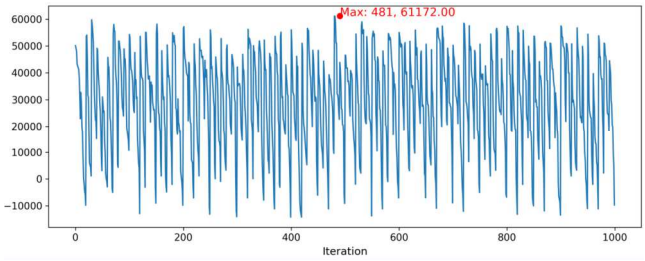

Fig. 4:Optimal value of energy

From the visualization results shown in Fig. 4, we can see that the final bank revenue determined by the pass rate and bad debt rate of the group 481 has reached the maximum value among the thousand groups, and the maximum value is 61,172 CNY, the accuracy of the simulated annealing process is confirmed. Group 481 can be traced back by the coding rule, representing the pass rate and bad debt rate at threshold 1 corresponding to the 49th card.

3 Credit Scoring Card Combination Optimization

Due to the complexity of banking scenarios, it is often necessary to use a combination of several different credit scoring cards to achieve an optimal risk control strategy. Next, we will study how to find the threshold combination corresponding to each scoring card given three credit scoring cards to maximize the bank's final income. The process of combining credit scoring cards is now simplified: if we choose the triple credit card combination strategy, the total pass rate after the combination of the three credit scoring cards is multiplied by the pass rates of all credit scoring cards, and the total bad debt rate is the average of the bad debt rates corresponding to the three credit scoring cards, thereby obtaining the corresponding loan interest income, bad debt losses and ultimately the bank's income.

The variables x_i, x_j, x_k are introduced in different value range of $i \in [1,10], j \in [11,20], k \in [21,30]$ to denote the selection among the 10 thresholds for credit scoring cards 1, 2 and 3 respectively, where three credit scoring cards and their corresponding thresholds are shown in the Table 1 below.

Table 1: Thresholds corresponding to the credit scoring card

	Credit Scoring Card 1		Credit Scoring Card 2		Credit Scoring Card 3	
Thres hold	Pass rate	Bad debt rate	Pass rate	Bad debt rate	Pass rate	Bad debt rate
1	0.76	0.013	0.72	0.032	0.8	0.012
2	0.77	0.015	0.73	0.038	0.82	0.013
3	0.78	0.017	0.76	0.05	0.83	0.024
4	0.8	0.024	0.77	0.053	0.86	0.04
5	0.82	0.026	0.79	0.065	0.89	0.042
6	0.84	0.028	0.82	0.074	0.92	0.057
7	0.87	0.03	0.86	0.076	0.93	0.068
8	0.93	0.036	0.87	0.079	0.94	0.069
9	0.94	0.043	0.91	0.08	0.97	0.07
10	0.96	0.052	0.92	0.084	0.98	0.073

Total pass rate:

$$A = a_i \times a_j \times a_k \tag{11}$$

Total bad debt rate:

$$B = \frac{1}{3} \times (b_i + b_j + b_k) \tag{12}$$

Interest income on loans:

$$W_1 = 100 \times 0.08 \times A \times (1 - B) \tag{13}$$

Bad debt losses:

$$W_2 = 100 \times A \times B \tag{14}$$

Final bank income:

$$\begin{aligned} W &= W_1 - W_2 = 8A \cdot (1-B) - 100A \cdot B \\ &= 8(a_i \times a_j \times a_k) - 36(a_i \times a_j \times a_k) \cdot (b_i + b_j + b_k) \end{aligned} \tag{15}$$

Seeking to combine $\{x_i, x_j, x_k\}$ credit scoring cards and their thresholds corresponding to different pass rates and bad debt rates to maximize the bank's final revenue, a 0-1 programming model is built to obtain the objective function:

$$\begin{aligned} maxW &= 8(\sum_{i=1}^{10} a_i x_i \cdot \sum_{j=11}^{20} a_j x_j \cdot \sum_{k=21}^{30} a_k x_k) \\ &-36(\sum_{i=1}^{10} a_i x_i \cdot \sum_{j=11}^{20} a_j x_j \cdot \sum_{k=21}^{30} a_k x_k) \\ &\times(\sum_{i=1}^{10} b_i x_i + \sum_{j=11}^{20} b_j x_j + \sum_{k=21}^{30} b_k x_k) \end{aligned} \tag{16}$$

For x_i, x_j, x_k only 0 or 1 can be taken and only one card can be taken from each of credit scoring card 1, credit scoring card 2 and credit scoring card 3, so using the nature of the 0-1 variable, the constraints of x_i, x_j, x_k are constrained to sum to 1 in the corresponding range of values, which leads to the following constraints:

$$s.t. \begin{cases} \sum_{i=1}^{10} x_i = 1 \\ \sum_{j=11}^{20} x_j = 1 \\ \sum_{k=21}^{30} x_k = 1 \\ x_i \in \{0,1\}, i = 1,2,...,10 \\ x_j \in \{0,1\}, j = 11,12,...,20 \\ x_k \in \{0,1\}, k = 21,22,...,30 \end{cases} \tag{17}$$

In order to convert the objective function into QUBO form, we need to decompose the quadratic and cross terms of the objective function into products between binary

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variables and each term in the objective function into quadratic and primary terms. Since the objective function contains cubic terms, replace the product of two variables with one variable by x_p instead of x_j, x_k, so the objective function is:

$$\begin{aligned} maxW &= 8(\sum_{i=1}^{10} a_i x_i \cdot \sum_{p=11}^{30} a_p x_p) \\ &-36(\sum_{i=1}^{10} a_i x_i \cdot \sum_{p=11}^{30} a_p x_p)\cdot(\sum_{i=1}^{10} b_i x_i + \sum_{p=11}^{30} b_p x_p) \end{aligned} \tag{18}$$

while satisfying the new constraints:

$$x_p = x_j x_k, \sum_{p=11}^{30} x_p = 1 \tag{19}$$

and $x_p = \begin{cases} 1, & \text{Select the j,k-th threshold} \\ 0, & \text{Don't select the j,k-th threshold} \end{cases}$

Add the constraint W' :

$$\begin{aligned} W' &= 8(a\sum_{i=1}^{10} a_i x_i \cdot b\sum_{p=11}^{30} a_p x_p) \\ &-36(a\sum_{i=1}^{10} a_i x_i \cdot b\sum_{p=11}^{30} a_p x_p)\cdot(a\sum_{i=1}^{10} b_i x_i + b\sum_{p=11}^{30} b_p x_p) \end{aligned} \tag{20}$$

The goal is to maximize the original objective function $maxW$ while satisfying the constraints, so that the constraint W' is also maximized when it is brought into the variable x_i, x_p that maximizes the objective $maxW$. Thus, the appropriate coefficient (a,b) is found so that when x_i, x_p satisfies $maxW = 0$, $W' = 0$, and when it does not, $W' > 0$.

By introducing a quadratic penalty objective function as an alternative to explicitly imposing constraints, the penalty introduced is understood as making the original constraints an effect of the solution process, which can be achieved by a natural function of the optimizer. The parameter P is a scalar penalty value that must be large enough to ensure that the penalty term is indeed equal to the constraint.

For the constraint $\sum_{i=1}^{10} x_i = 1$, we introduce an equivalent quadratic penalty function $P_{equivalent} = (1-\sum_{i=1}^{10} x_i)^2$.

Expanding it:

$$P_{equivalent} = (1 - x_1 - ... - x_{10})^2 = 1 - 2\sum_{i=1}^{10} x_i + (\sum_{i=1}^{10} x_i)^2 \tag{21}$$

Since x_i is a binary variable, $x_i^2 = x_i$, after bringing it in, we can get $P_{equivalent} = 1 - \sum_{i=1}^{10} x_i + \sum_{i<l} x_i x_l$, where x_l is the term when $i < l$, the equivalent penalty function for the other constraints can be obtained in the same way.

Table 2: Equivalent penalty function

restrictive condition	Equivalent punishment
$\sum_{i=1}^{10} x_i = 1$	$P(1-\sum_{i=1}^{10} x_i + \sum_{i<l} x_i x_l)$
$\sum_{p=11}^{30} x_p = 1$	$P(1-\sum_{p=11}^{30} x_p + \sum_{p<l} x_p x_l)$

Thus, the objective function can be written as:

$$\begin{aligned} maxW &= 8(\sum_{i=1}^{10} a_i x_i \cdot \sum_{p=11}^{30} a_p x_p) \\ &-36(\sum_{i=1}^{10} a_i x_i \cdot \sum_{p=11}^{30} a_p x_p)\cdot(\sum_{i=1}^{10} b_i x_i + \sum_{p=11}^{30} b_p x_p) \\ &-P[(1-\sum_{i=1}^{10} x_i + \sum_{i<l} x_i x_l) + (1-\sum_{p=11}^{30} x_p + \sum_{p<l} x_p x_l)] \end{aligned} \tag{22}$$

where P is a positive scalar and the problem is unconstrained when P is sufficiently large.

Due to the presence of the cubic term therefore another variable substitution $x_q = x_i x_p = x_i x_j x_k$ is performed while satisfying the new constraints:

$$\sum_{q=1}^{30} x_q = 1, x_q = \begin{cases} 1, & \text{Select the i,j,k-th threshold} \\ 0, & \text{Don't select the i,j,k-th threshold} \end{cases}$$

Since the QUBO model solves the minimum value optimization problem, the objective function is:

$$\begin{aligned} maxW = min(-H(x)) &= 8(\sum_{q=1}^{30} a_q x_q) \\ &-36(\sum_{q=1}^{30} a_q x_q)\cdot(\sum_{q=1}^{30} b_q x_q) - P(1-\sum_{q=1}^{30} x_i + \sum_{q<l} x_q x_l) \end{aligned} \tag{23}$$

The QUBO form is obtained, collapsed into the general form as:

$$\begin{aligned} min(-H(x)) &= 8\sum_{q=1}^{30} a_q x_q + P\sum_{q=1}^{30} x_q \\ &-36(\sum_{q=1}^{30}\sum_{q=1}^{30} a_q x_q b_q x_q) - P\sum_{q<l} x_q x_l - P \end{aligned} \tag{24}$$

Thus define X to be a column vector of 30×1 where three elements take the value of 1 and the remaining elements take the value of 0:

$X = (x_1, \cdots, x_{10}, \quad x_{11}, \cdots, x_{20}, \quad x_{21}, \cdots, x_{30})^T$, and Q to be the matrix of 30×30 with the corresponding elements being the coefficients of the function formed by the pass rate and the bad debt rate corresponding to the credit scoring cards 1, 2, and 3 and their thresholds.

The D-Wave quantum annealing algorithm can solve a small preliminary mixed integer optimization problem with 16 binary variables and 16 constraints [10-11], thereby creating a transformation that increases the number of QUBO variables but maintains the number of cross product terms , therefore, the quantum annealing process is iterated until the optimal solution is found with a high enough probability. The quantum annealer explores all possible inputs in parallel to find the best solution to the problem. Using this algorithm to solve the problem contributes to the realization of quantum computers. At the same time, we delve into the use of quantum approximate optimization algorithms to better solve approximate solutions to combination optimization problems, where the potential advantages of larger values of parameter P may be offset by the reduction in accuracy.

The quantum annealing process can be described by the

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Hamiltonian $H(s)$, which is an operator related to the total energy of the quantum system, encoding the matrix into the form of a Hamiltonian, which in turn acts on the quantum state, and Hamiltonian simulation involves the efficient approximation of the target Hamiltonian evolution through the implementation of a unitary evolution

$$H(s) = A(s)H_I - B(s)H_P \tag{25}$$

where $A(s)$ and $B(s)$ are annealing path functions defined according to the normalized annealing time $s = \frac{t}{t_a}$ initially at $A(s) = 1, B(s) = 0$ and after annealing at $A(1) = 0, B(1) = 1$.

The pyqubo library in Python can be used in this problem can simulate the process of quantum annealing algorithm to solve the obtained:

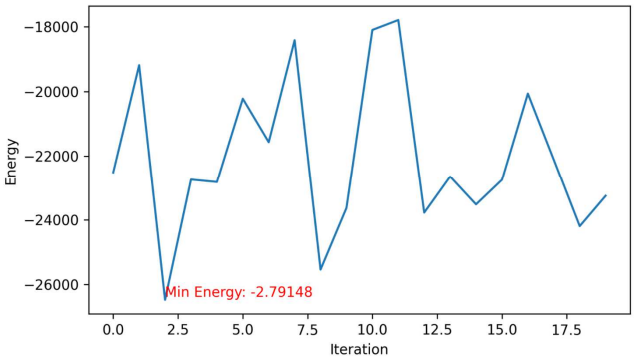

Fig. 5:Simulated annealing process

As shown in Fig. 5, after identifying the three credit scoring cards we combine their thresholds, and the final result of combining the decision variables with the objective function of maximum revenue is the combination of the eighth threshold for the first card corresponds to a pass rate and bad debt rate of 0.87 and 0.03, the first threshold for the second card corresponds to a pass rate and bad debt rate of 0.72 and 0.032, and the second threshold for the third card corresponds to a pass rate and bad debt rate of 0.82 and 0.013. This brings the bank's final revenue to the maximum value of 27,914.8 CNY.

4 Conclusions

In this paper, a QUBO model based on credit scoring card threshold combination optimization is developed. By exploring the single credit scoring card threshold optimization problem, an optimization model is established based on the functional relationship that determines bank revenue, and transformed into QUBO form, the simulated annealing algorithm is used to find the 49th threshold 1 out of 100 credit scoring cards to maximize the bank's final revenue. Then, after conducting modeling research on the combination of credit scoring card thresholds, a QUBO model is constructed through processes such as variable substitution and penalty function addition, and the quantum annealing algorithm is used to finally determine the threshold combination corresponding to the three credit scoring cards and obtain the bank maximum income.

REFERENCES

[1] Xu N, Yuan Y, Rong Z, Depressed access to formal finance and the use of credit card debt in Chinese SMEs, *China Economic Review*, 72: 101758, 2022.

[2] Drumond I, Jorge J, Loan interest rates under risk-based capital requirements: The impact of banking market structure, *Economic Modelling*, 32: 602-607, 2013.

[3] Fatouh M, Giansante S, The cyclicality of bank credit losses and capital ratios under expected loss model, *Annals of Operations Research*, 330(1-2):807-840, 2023.

[4] Sohn S Y, Lim K T, Ju Y, Optimization strategy of credit line management for credit card business, *Computers and operations research*, 48: 81-88, 2014.

[5] Pasricha P, Selvamuthu D, D'Amico G, et al, Portfolio optimization of credit risky bonds: a semi-Markov process approach, *Financial Innovation*, 6(1): 1-14, 2020.

[6] Verma A, Lewis M, Goal Seeking Quadratic Unconstrained Binary Optimization, *Results in Control and Optimization*, 7: 100125, 2022.

[7] Pisani L, Moreau G S, Leonardi E, et al, Multi-tower heliostat field optimization by means of adiabatic quantum computer, *Solar Energy*, 263: 111893, 2023.

[8] Hadfield S, Wang Z, O'gorman B, et al, From the quantum approximate optimization algorithm to a quantum alternating operator ansatz, *Algorithms*, 12(2): 34, 2019.

[9] Lucas A, Ising formulations of many NP problems, *Frontiers in physics*, 2: 74887, 2014.

[10] Boettcher S, Extremal Optimization for Quadratic Unconstrained Binary Problems, *Physics Procedia*, 68: 16-19, 2015.

[11] Künnemann H, Phillipson F, Finding Debt Cycles: QUBO Formulations for the Maximum Weighted Cycle Problem Solved Using Quantum Annealing, *Mathematics*, 11(12): 2741, 2023.

[12] Alnowibet K A, Mahdi S, El-Alem M, et al, Guided hybrid modified simulated annealing algorithm for solving constrained global optimization problems, *Mathematics*, 10(8): 1312, 2022.

[13] Willsch D, Willsch M, Gonzalez Calaza C D, et al, Benchmarking Advantage and D-Wave 2000Q quantum annealers with exact cover problems, *Quantum Information Processing*, 21(4): 141, 2022.

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